

REPORT SUBMISSION

Experimental Verification of Neutral Point Location for a Wing–Tail Configuration Using Glide Tests

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Github repository of Project

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Abstract

This report presents a longitudinal steady-flight trim analysis and a corresponding performance evaluation in terms of thrust required and power required for a small fixed-wing UAV. Using a classical aerodynamic model consisting of a linear lift curve and a parabolic drag polar, trim elevator deflection is computed by enforcing zero pitching moment about the center of gravity at each operating angle of attack. For each trimmed state, the steady-level flight speed is obtained from the lift equilibrium, and thrust and power requirements are derived from the resulting drag. The computed results show physically consistent trends and identify the operating points corresponding to minimum thrust required and minimum power required, which are critical for range and endurance respectively.

Chapter 1

Introduction

Longitudinal trim and steady-flight performance analysis are foundational tasks in fixed-wing aircraft design because they quantify whether an aircraft can sustain a prescribed flight condition without acceleration or continuous control correction. For a UAV operating in steady, level flight, a trimmed state is achieved when the forces and moments about the center of gravity are in equilibrium: the net pitching moment is zero, the lift force balances the weight, and the propulsive thrust balances the aerodynamic drag. Establishing these equilibrium conditions is essential for assessing control authority, defining feasible operating points, and providing reliable inputs for propulsion sizing and mission planning.

In this project, the UAV is analyzed across a sweep of angle of attack, α , which serves as the primary parameter governing lift generation and aerodynamic loading. For each value of α , the required elevator deflection, $\delta_{e,\text{trim}}$, is computed by enforcing the pitching-moment trim condition $C_m = 0$. Once the aircraft is trimmed, the corresponding steady level-flight speed V is obtained from the lift equilibrium $L = W$. Using this trimmed speed, the thrust required T_r is computed from the drag force, and the power required P_r is evaluated as $P_r = T_r V$. This workflow produces a consistent set of trim and performance quantities over the operating envelope.

The resulting thrust- and power-required characteristics are then used to identify key performance points. Specifically, the minimum thrust required condition corresponds to minimum drag and typically aligns with maximum aerodynamic efficiency (L/D), while the minimum power required condition corresponds to the most energy-efficient steady flight and is directly relevant for maximum endurance in propeller-driven aircraft. These operating points are of practical importance for selecting cruise and loiter speeds and for defining propulsion-system margins.

The remainder of this report is organized as follows. Chapter 2 presents the aircraft geometry derivations, aerodynamic and trim relations, and the numerical methodology. Chapter 3 reports the computed results using figures and tables. Chapter 4 discusses the physical interpretation and design implications of the observed trends. Finally, Chapter 5 summarizes the key findings and highlights the most relevant operating conditions for performance and mission objectives.

Chapter 2

Theory and Method

This chapter describes the theoretical foundations and the computational procedure used to perform the trim and performance analysis. The formulation is based on classical fixed-wing aerodynamic theory and steady, level-flight assumptions, which are well suited for preliminary design and performance evaluation of small UAVs.

2.1 Wing Geometry and Derived Parameters

The wing geometry directly influences lift generation, induced drag, and overall aerodynamic efficiency. One of the key geometric descriptors is the wing taper ratio, defined as

$$\lambda = \frac{c_t}{c_r}, \quad (2.1)$$

where c_t and c_r are the wing tip and root chord lengths, respectively. For the given UAV configuration, the computed taper ratio is $\lambda = 0.1667$, indicating a strongly tapered wing planform.

The wing reference area is obtained using the trapezoidal planform relation

$$S = \frac{b}{2} (c_r + c_t), \quad (2.2)$$

which yields a total wing area of $S = 0.7875 \text{ m}^2$. This area is used as the reference for all aerodynamic force and coefficient calculations.

The aspect ratio, defined as

$$AR = \frac{b^2}{S}, \quad (2.3)$$

is a critical parameter governing induced drag. For the present design, $AR = 2.8571$, which is typical of compact UAV configurations where structural and packaging constraints limit wingspan.

Induced drag is incorporated through a parabolic drag polar using the induced drag factor

$$k = \frac{1}{\pi e AR}, \quad (2.4)$$

where e is the Oswald efficiency factor. Based on the selected efficiency value and the computed aspect ratio, the induced drag factor is $k = 0.125178$. This parameter quantifies the penalty associated with lift generation and becomes increasingly significant at higher angles of attack.

2.2 Aerodynamic Model

The aerodynamic forces are modeled using a simplified, yet widely accepted, linear lift and quadratic drag formulation. The lift coefficient is expressed as

$$C_L = C_{L0} + C_{L\alpha}\alpha, \quad (2.5)$$

where C_{L0} represents the lift coefficient at zero angle of attack and $C_{L\alpha}$ denotes the lift-curve slope. This linear approximation is valid within the pre-stall operating range considered in this study.

The drag coefficient is modeled using a parabolic drag polar:

$$C_D = C_{D0} + kC_L^2, \quad (2.6)$$

where C_{D0} accounts for parasitic drag components such as skin friction and form drag, and the term kC_L^2 represents induced drag. This formulation captures the rapid growth of drag at higher lift levels and forms the basis for thrust and power requirement calculations.

2.3 Trim Condition (Pitching Moment Equilibrium)

Longitudinal trim is enforced by requiring that the net pitching moment coefficient about the aircraft center of gravity be zero. The pitching moment coefficient is modeled as a linear function of angle of attack and elevator deflection:

$$C_m = C_{m0} + C_{m\alpha}\alpha + C_{m\delta_e}\delta_e. \quad (2.7)$$

The trim condition is obtained by setting $C_m = 0$, which yields the required elevator deflection for a given angle of attack:

$$\delta_{e,\text{trim}} = -\frac{C_{m0} + C_{m\alpha}\alpha}{C_{m\delta_e}}. \quad (2.8)$$

This relation is evaluated for each α in the sweep, ensuring that all subsequent performance calculations correspond to a longitudinally trimmed state.

2.4 Steady Level Flight: Speed, Thrust Required, and Power Required

Under steady, level-flight conditions, the lift force balances the aircraft weight:

$$L = W = \frac{1}{2}\rho V^2 SC_L, \quad (2.9)$$

where ρ is the air density and V is the flight speed. Solving this equation for the speed gives

$$V = \sqrt{\frac{2W}{\rho SC_L}}. \quad (2.10)$$

The aerodynamic drag is computed as

$$D = \frac{1}{2}\rho V^2 SC_D. \quad (2.11)$$

In steady flight, the thrust required T_r must balance the drag force, such that

$$T_r = D. \quad (2.12)$$

The power required to maintain steady flight is then obtained from the product of thrust required and flight speed:

$$P_r = T_r V. \quad (2.13)$$

This quantity directly represents the mechanical power that must be delivered by the propulsion system, excluding propulsive efficiency losses.

2.5 Computational Workflow

The complete analysis is carried out numerically by performing a parametric sweep over angle of attack α within the specified operating range. For each value of α , the following steps are executed in sequence:

- the trim elevator deflection $\delta_{e,\text{trim}}$ is computed from the pitching-moment equilibrium condition,
- the lift and drag coefficients $C_L(\alpha)$ and $C_D(\alpha)$ are evaluated using the aerodynamic model,
- the steady level-flight speed V is obtained from the lift balance,
- the thrust required T_r and power required P_r are calculated,
- additional performance metrics, such as C_L/C_D , are derived for efficiency assessment.

All computed quantities are stored and exported to a CSV file for traceability and post-processing, and all plots used in this report are generated directly from the same dataset to ensure consistency and reproducibility.

Chapter 3

Results

This chapter presents the numerical results obtained from the trim and performance analysis. All results are generated under steady, level-flight assumptions and are based on the aerodynamic and trim models described in Chapter 2. The section is organized to first summarize the derived geometry, followed by trim behavior, aerodynamic characteristics, thrust and power requirements, efficiency metrics, and key operating points.

3.1 Geometry Summary

The derived geometric parameters form the foundation for all aerodynamic and performance calculations. Table 3.1 summarizes the wing geometry and the induced-drag factor used throughout the analysis.

Table 3.1: Derived geometric and induced-drag parameters.

Parameter	Value
Taper ratio, λ	0.1667
Wing area, S [m ²]	0.7875
Aspect ratio, AR	2.8571
Induced drag factor, k	0.125 178

The relatively low aspect ratio indicates a compact wing configuration, which naturally increases the contribution of induced drag at higher lift coefficients. This effect is reflected in the subsequent drag, thrust, and power characteristics.

3.2 Trim Behavior

Figure 3.1 illustrates the elevator deflection required to maintain longitudinal trim as a function of angle of attack. As the angle of attack increases, the aerodynamic pitching moment contribution changes, requiring progressively larger elevator deflections to satisfy the pitching-moment equilibrium condition ($C_m = 0$). The smooth and nearly linear trend confirms that the aircraft remains within the linear control regime over the analyzed range.

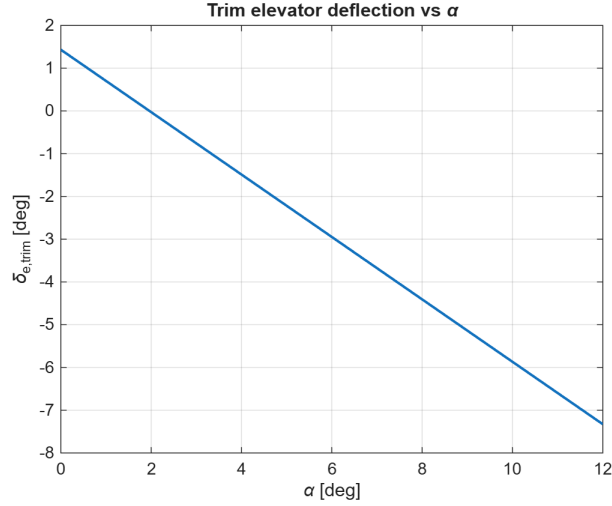


Figure 3.1: Trim elevator deflection $\delta_{e,trim}$ versus angle of attack.

Figure 3.2 shows the corresponding trimmed steady-flight speed as a function of angle of attack. Increasing α leads to a higher lift coefficient, thereby reducing the speed required to satisfy the lift equilibrium $L = W$. This inverse relationship between angle of attack and trimmed speed is consistent with classical steady-flight theory.

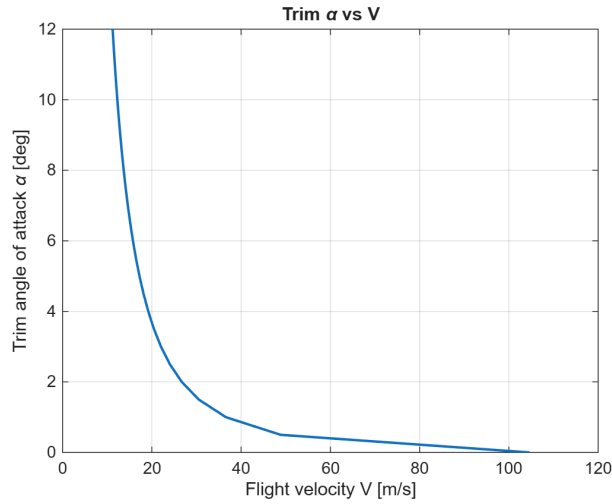


Figure 3.2: Trimmed steady-flight speed V versus angle of attack α .

3.3 Aerodynamic Coefficients

The variation of aerodynamic coefficients with angle of attack is shown in Figures 3.3 and 3.4. The lift coefficient exhibits a linear increase with angle of attack, validating the assumed linear lift model within the operating range considered.

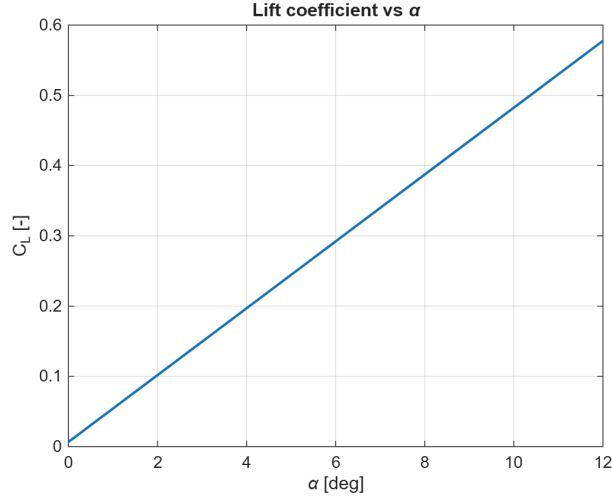


Figure 3.3: Lift coefficient C_L versus angle of attack.

The drag coefficient increases nonlinearly with angle of attack, as shown in Figure 3.4. This convex trend arises from the quadratic induced-drag term kC_L^2 , which becomes increasingly dominant at higher lift coefficients.

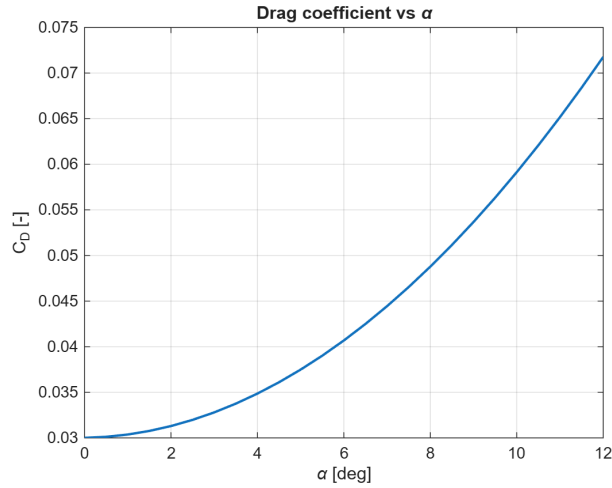


Figure 3.4: Drag coefficient C_D versus angle of attack.

3.4 Thrust Required and Power Required

The thrust required as a function of flight speed is presented in Figure 3.5. The curve exhibits the characteristic U-shaped profile, reflecting the trade-off between induced drag at low speeds and parasitic drag at high speeds. The minimum point on this curve corresponds to the condition of minimum aerodynamic drag.

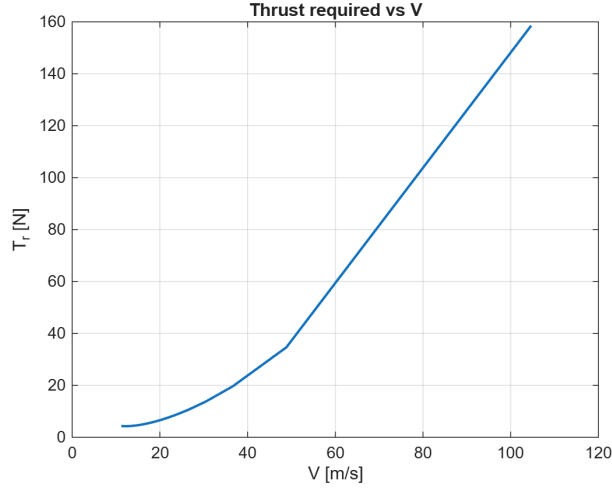


Figure 3.5: Thrust required T_r versus flight speed V .

Figure 3.6 shows the power required as a function of flight speed. The minimum of the power curve occurs at a lower speed than the minimum thrust point, which is a well-known characteristic of propeller-driven aircraft. This operating condition corresponds to maximum endurance.

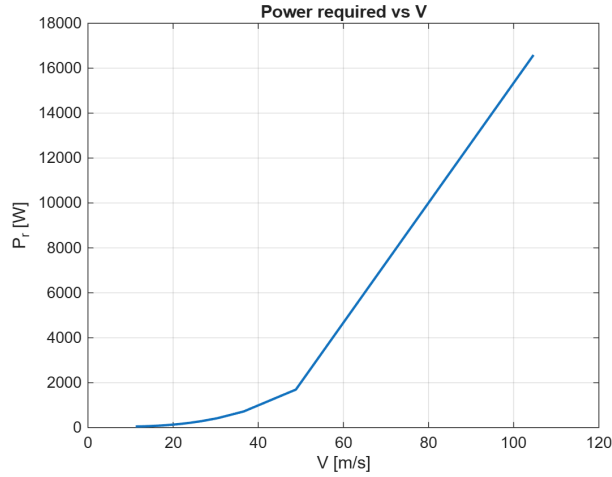


Figure 3.6: Power required P_r versus flight speed V .

3.5 Efficiency Metrics

Aerodynamic efficiency is evaluated using the lift-to-drag ratio, C_L/C_D , shown in Figure 3.7. The peak of this curve corresponds to the most aerodynamically efficient operating condition and aligns closely with the minimum thrust required point.

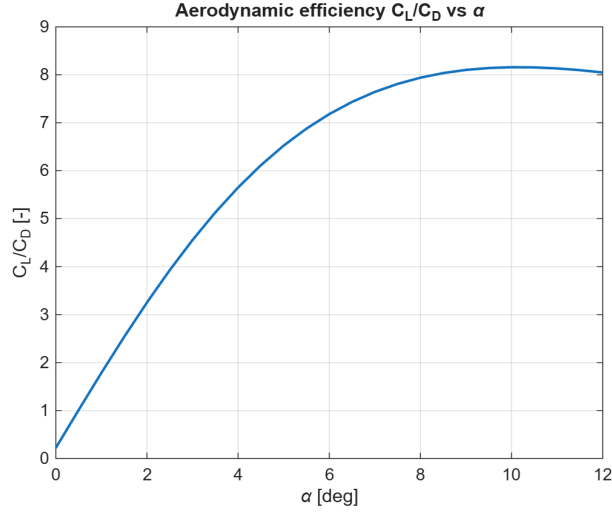


Figure 3.7: Aerodynamic efficiency C_L/C_D versus angle of attack.

Figure 3.8 presents the endurance-related metric $C_L^{3/2}/C_D$, which is commonly used to identify the minimum power condition. The maximum of this curve occurs near the angle of attack associated with minimum power required.

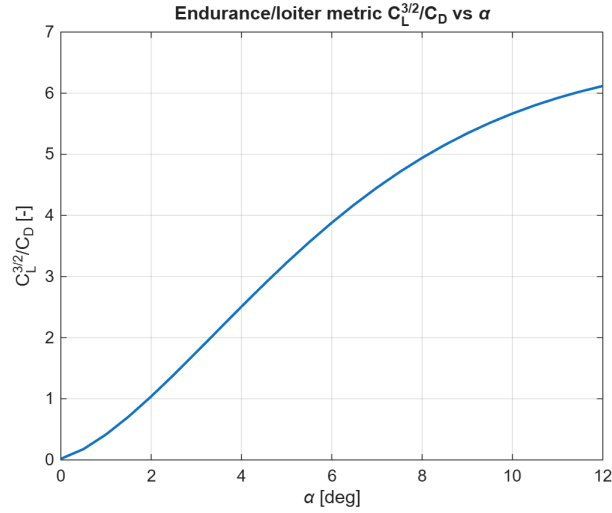


Figure 3.8: Endurance-related metric $C_L^{3/2}/C_D$ versus angle of attack.

3.6 Key Operating Points

The most important operating points extracted from the trim and performance sweep are summarized in Table 3.2. These points are directly relevant for mission planning and propulsion-system sizing.

Table 3.2: Key operating points from the computed trim and performance analysis.

Condition	α [deg]	V [m/s]	T_r [N]	P_r [W]
Minimum thrust required ($T_{r,\min}$)	10.0	12.263	4.290	52.611
Minimum power required ($P_{r,\min}$)	12.0	11.208	4.349	48.737

3.7 Representative Data Snapshot

To demonstrate numerical consistency and provide traceability, a representative subset of the computed results is presented in Table 3.3. The full dataset is available in the exported CSV file and is referenced in the appendix.

Table 3.3: Representative trimmed steady-flight states from the CSV results.

α [deg]	$\delta_{e,\text{trim}}$ [deg]	C_L	C_D	V [m/s]	T_r [N]	P_r [W]	C_L/C_D
2	−0.028	0.101 800	0.031 297	26.698	10.760	287.284	3.253
4	−1.488	0.196 974	0.034 857	19.193	6.194	118.877	5.651
6	−2.948	0.292 149	0.040 684	15.760	4.874	76.814	7.181
8	−4.408	0.387 323	0.048 779	13.687	4.408	60.332	7.940
10	−5.868	0.482 498	0.059 142	12.263	4.290	52.611	8.158
12	−7.328	0.577 672	0.071 773	11.208	4.349	48.737	8.049

Chapter 4

Discussion

This chapter interprets the numerical results presented in Chapter 3 from a physical and engineering perspective. The objective is to relate the observed trends in trim, speed, thrust, and power to fundamental aerodynamic principles and to highlight their implications for aircraft performance and design.

4.1 Trim Trends and Longitudinal Equilibrium

The trim curve shown in Figure 3.1 illustrates how the elevator deflection must vary to maintain longitudinal equilibrium as the angle of attack changes. As α increases, the aerodynamic pitching moment contribution associated with the angle of attack term $C_{m\alpha}\alpha$ increases linearly. To satisfy the trim condition $C_m = 0$, the elevator-generated pitching moment through the term $C_{m\delta_e}\delta_e$ must counteract this change.

The resulting near-linear relationship between $\delta_{e,\text{trim}}$ and α is consistent with the assumed linear pitching-moment model and confirms that the aircraft remains within the linear control regime across the analyzed range. This behavior indicates predictable longitudinal control characteristics and adequate elevator authority for steady-flight operation.

4.2 Speed Reduction with Increasing Angle of Attack

Figure 3.2 shows a monotonic decrease in trimmed steady-flight speed with increasing angle of attack. This trend follows directly from the lift equilibrium relation

$$V = \sqrt{\frac{2W}{\rho S C_L}}.$$

As the angle of attack increases, the lift coefficient C_L increases, allowing the required lift to be generated at a lower dynamic pressure and therefore at a lower flight speed.

The smooth and physically consistent variation of speed with angle of attack serves as an important internal validation of both the aerodynamic model and the numerical solver. It also highlights the practical trade-off between operating at higher angles of attack and reduced flight speed.

4.3 Separation of Minimum Thrust and Minimum Power Conditions

The condition of minimum thrust required corresponds to minimum aerodynamic drag, which is achieved at the angle of attack that maximizes the lift-to-drag ratio C_L/C_D . This relationship is evident in Figure 3.7, where the peak aerodynamic efficiency occurs at $\alpha = 10^\circ$. This value coincides with the minimum thrust required operating point reported in Table 3.2, where the UAV achieves

$$T_{r,\min} = 4.290 \text{ N} \quad \text{at} \quad \alpha = 10^\circ, V = 12.263 \text{ m/s.}$$

In contrast, the minimum power required condition occurs at a slightly higher angle of attack and a lower flight speed. Since power required is given by $P_r = T_r V$, a modest increase in thrust at lower speed can be offset by a sufficient reduction in velocity, resulting in a lower overall power demand. This effect explains why the minimum power point is found at

$$P_{r,\min} = 48.737 \text{ W} \quad \text{at} \quad \alpha = 12^\circ, V = 11.208 \text{ m/s.}$$

This operating condition is of particular importance for endurance-oriented missions, such as loiter or surveillance, where minimizing energy consumption is critical.

4.4 Engineering Relevance for Propulsion Sizing and Mission Planning

From a propulsion-system perspective, the thrust-required and power-required curves shown in Figures 3.5 and 3.6 provide direct quantitative inputs for motor and propeller selection. The propulsion system must be capable of delivering thrust and power levels that exceed the steady-flight requirements with adequate margin.

In practice, this margin must account for propulsive efficiency losses, climb and acceleration demands, atmospheric disturbances, and control authority requirements. Operating points corresponding to minimum thrust and minimum power also serve as reference conditions for defining cruise and loiter speeds, respectively, thereby linking aerodynamic performance directly to mission-level decision making.

Chapter 5

Conclusion

This report presented a comprehensive longitudinal trim and steady-flight performance analysis for a fixed-wing UAV using classical aerodynamic modeling and steady, level-flight assumptions. The analysis was carried out over a prescribed angle-of-attack sweep and employed a linear lift model, a parabolic drag polar, and a linear pitching-moment formulation to ensure analytical clarity and numerical robustness.

From the aircraft geometry, the key derived parameters were obtained as $\lambda = 0.1667$, $S = 0.7875 \text{ m}^2$, $AR = 2.8571$, and $k = 0.125178$. These parameters were used consistently throughout the trim and performance calculations. The resulting trends in trim elevator deflection, lift and drag coefficients, trimmed flight speed, thrust required, and power required were found to be smooth and physically consistent, providing confidence in both the modeling assumptions and the numerical implementation.

Two performance-critical operating points were identified from the computed results. The minimum thrust required condition, $T_{r,\min} = 4.290 \text{ N}$ at $\alpha = 10^\circ$ and $V = 12.263 \text{ m/s}$, corresponds to minimum aerodynamic drag and maximum lift-to-drag ratio, and is therefore most relevant for range-efficient operation. The minimum power required condition, $P_{r,\min} = 48.737 \text{ W}$ at $\alpha = 12^\circ$ and $V = 11.208 \text{ m/s}$, corresponds to the most energy-efficient steady-flight state and is directly associated with maximum endurance for a propeller-driven UAV.

Overall, the identified trim and performance characteristics provide clear and quantitative guidance for mission planning, including the selection of cruise and loiter speeds, as well as for propulsion-system sizing and performance margin assessment. The methodology adopted in this work can be readily extended to include propulsive efficiency, climb performance, or dynamic stability analysis in future studies.

Appendix A

Full Data and Reproducibility Note

To ensure transparency, traceability, and reproducibility of the results presented in this report, all numerical outputs generated during the trim and performance analysis are archived and made available in machine-readable form. The complete set of computed data for the entire angle-of-attack sweep is exported to the CSV file `project4_results.csv`, which includes trim elevator deflection, aerodynamic coefficients, trimmed flight speed, thrust required, power required, and derived efficiency metrics for each operating point.

For clarity and conciseness, only a representative subset of this dataset is presented in tabular form in Chapter 3. The full CSV file may be used for independent verification of the reported results, additional post-processing, or the generation of alternative plots without the need to rerun the original simulations.

All figures referenced throughout this report are generated directly from the same computational run and are stored in the `figures/` directory using consistent and descriptive file naming. This organization ensures that the figures can be unambiguously traced back to the underlying numerical data and that the entire analysis can be reproduced by executing the provided code with the same input parameters.